

Subject Description Form

Subject Code	ABCT3391
Subject Title	Ecology and Ecosystem
Credit Value	3
Level	3
Pre-requisite	CSE29371 Environmental Chemistry
Objectives	The subject explains basic principles and concepts of ecosystem ecology. The relationships between organisms and the environment will be taught. Apart from the biotic-abiotic interactions, students will learn about the human-induced changes in ecosystems, analytical skills to address environmental problems and possible remediation methods.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. know the diversity of ecosystems in the world; b. better understand the physicochemical and biological structures of characteristic ecosystems; c. aware of the severity of environmental deterioration; d. define ecology and different levels of biological organization; e. describe trophic levels and the fate of energy as it passes from one level to another; and f. apply their knowledge of ecology and environmental science to improve the quality of life in the individual context and as citizens in a global village.
Subject Synopsis/ Indicative Syllabus	1. <u>Concepts of ecology</u> Evolution; natural selection; biodiversity; extinction and speciation; levels of biological organization 2. <u>Communities and ecosystems</u> Physical environments; organism interactions; ecosystem functions and services; characteristics of key ecosystems 3. <u>Environmental mechanisms</u> The water cycle; trophic transfer of energy; production and decomposition; nutrient cycling 4. <u>Ecological monitoring and conservation</u> Anthropogenic stress; environmental deterioration; ecological restoration; ecological assessment methods
Teaching/Learning Methodology	Interactive lectures will facilitate students' learning of key concepts of environmental science and ecology, and promote communication between teachers and students. Tutorials will include small group discussion of relevant topics to enhance knowledge sharing among students and their awareness of recent environmental and ecological issues. Group projects will encourage students to work as a team, to research into interesting environmental and ecological topics, and to present their findings in class, and prepare written report. Assignments will be arranged to support students' flipped classroom learning. If possible, a virtual or real field trip will be organised for students to experience the ecology of Hong Kong.

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed					
			a	b	c	d	e	f
	1. Assignment	30%					✓	✓
	2. Test	30%	✓	✓	✓	✓		
	3. Examination	40%	✓	✓	✓	✓	✓	✓
	Total	100%						
	<u>Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:</u> The assignment aims to foster students' interests in ecology and environmental conservation. Through information searching and discussion on selected topics, students will gain better insights into real-world environmental and ecological issues. The test and examination will evaluate students' level of understanding of the taught topics. High-order thinking questions will be used to assess students' analytical and problem-solving skills.							
Student Study Effort Required	Class contact:							
	• Lecture • Tutorial						27 hours	
	Other study effort:							
	• Assignment, test and self-study (and a field trip, if any)						81 hours	
	Total study effort:						120 hours	
Reading List and References	1. Smith TM and Smith RL (2015) <i>Elements of Ecology</i>, 9th ed. Pearson. QH541.S62 2015 2. Botkin DB and Keller EA (2014) <i>Environmental Science: Earth as a Living Planet</i>, 9th ed. Wiley. GE105.B68 2014 3. Stiling PD (2002) <i>Ecology: theories and applications</i>, 4th ed. Prentice Hall. QH541 .S675 2002							